

Shan Yang

Staff Applied Scientist — Foundation Models, Multimodal AI, Generative AI

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PROFESSIONAL SUMMARY

- Technical leader with 8+ years driving foundation model research and production ML systems at scale across multimodal generative AI, video/image generation, and LLM post-training
- Proven track record managing Applied Scientists and establishing technical roadmaps, shipping products serving millions with \$500M+ revenue impact
- Deep expertise in multimodal foundation models: 3,500+ citations at premier venues (NeurIPS, ICCV, ECCV, ICML, AAI), pioneering novel architectures in attention-based fusion, music-conditioned motion synthesis, and autoregressive diffusion

TECHNICAL SKILLS

ML Frameworks: PyTorch, TensorFlow, Scikit-learn — Distributed training (DeepSpeed, FSDP)

Foundation Models: Diffusion Models, Multimodal LLMs, Transformers, CLIP, BLIP

Research Focus: Novel Model Architectures, Image/Video/3D Generation & Editing, Multimodal Understanding, Post-Training (RLHF), Test-Time Training, Reinforcement Learning

Architecture Innovation: Attention Bottlenecks, Multimodal Fusion, Music-Conditioned Motion Synthesis, Autoregressive Diffusion, Linear Recurrent Models

ENGINEERING & RESEARCH EXPERIENCE

Adobe Foundry

San Jose, CA

Staff Applied Scientist

March 2026 – Present

- Driving post-training research for Adobe Foundry, Adobe's family of generative AI foundation models powering Firefly and creative tooling at scale
- Developing post-training techniques (SFT, RLHF/DPO, LoRA, distillation, test-time training) to align foundation models with creator intent and production quality bars

Amazon Prime Video

Sunnyvale, CA

Senior Applied Scientist Tech Lead

June 2025 – March 2026

- Led GenAI Live Action Studio with **5 Applied Scientists**, building autoregressive diffusion architectures and training infrastructure for 10B+ parameter video generation models
- Advanced controllable generation via novel post-training techniques (Test-Time Training, Teacher-Diffusion-Forcing, RL), translating director and creative-team requirements into production-ready architectures

Amazon Search

Palo Alto, CA

Senior Applied Scientist Tech Lead

September 2021 – June 2025

- Led **Amazon Video Search** from zero to launch serving millions with **\$500M+ annual revenue gain**, managing **3 Applied Scientists and 2 MLEs** and influencing a 15-person SDE team; owned multi-year roadmap, multimodal ranker design, and production ML systems
- Pioneered multimodal fusion architecture integrating video, text, image, and structured data, enabling multi-modal search at scale
- Drove multimodal LLM robustness research: Schema-of-Thought framework achieving **+28% hallucination reduction**; published at ECCV, AAI and filed multiple patents

Google Research

Mountain View, CA

Senior Research Software Engineer

June 2017 – September 2021

- Led foundational research in multimodal generative models, influencing **3 Research Engineers**, driving research agenda in music-conditioned 3D motion synthesis and video-audio-motion modeling, publishing at premier ML conferences (ICCV, NeurIPS, ECCV)
- Drove technical ownership of 2D/3D pose reconstruction models deployed on GCP at scale, demonstrating ability to lead end-to-end ML product development from research to production
- Provided technical leadership through patent contributions, cross-functional collaboration, and mentorship of research interns on advanced ML projects

RESEARCH IN PROGRESS

Schema-of-Thought: Prompting framework for multimodal LLM robustness and out-of-distribution generalization (+28% Claude 3.5, +13.3% GPT-4o)

Teacher Diffusion Forcing: Novel training paradigm merging teacher forcing with diffusion forcing for autoregressive diffusion, modeling error as conditional distribution

ClipDreamer: Subject-driven video generation using test-time training to inject long video clip information for improved subject fidelity

SELECTED PUBLICATIONS

Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning

S Yang

arXiv 2026 (2605.14040)

Sole-author preprint. Three-stage corpus audit (Jaccard $\rightarrow$ embedding $\rightarrow$ LLM-judge), novel-source held-out olympiad benchmark (PhysOlym-A, 500q), and GSPO+DAPO recipe lifting Qwen3-VL-8B by +18.3 pp on PhysOlym-A.

Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control

P Wang, **S Yang**, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal

ICML 2026

Linear recurrent architecture leveraging optimal control theory for LLM test-time reasoning (+35.8% Math-500, +10% AIME25).

ViLA: Efficient Video-Language Alignment for Video Question Answering

X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, **S Yang**

ECCV 2024

Efficient vision-language multimodal foundation models.

ICAR: Image-based Complementary Auto Reasoning

X Wang, A Liang, J Liang, M Lin, Y Lou, **S Yang**

AAAI 2024

Multimodal reasoning for visual understanding tasks.

Attention Bottlenecks for Multimodal Fusion

A Nagrani, **S Yang**, A Arnab, A Jansen, C Schmid, C Sun

NeurIPS 2021 — **800+ citations**

Multimodal fusion architecture for foundation models.

AI Choreographer: Music Conditioned 3D Dance Generation with AIST++

R Li*, **S Yang***, DA Ross, A Kanazawa

ICCV 2021 — **700+ citations**

Multimodal generative modeling: music, motion, vision.

Physics-Inspired Garment Recovery from a Single-View Image

S Yang, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin

ACM ToG 2018 — **100+ citations**

3D generation from images.

Modeling Context in Referring Expressions

L Yu, P Poirson, **S Yang**, A Berg, T Berg

ECCV 2016 — **1600+ citations**

Vision-language understanding.

PATENTS & INTELLECTUAL PROPERTY

VLAP: Efficient Video-Language Alignment via Frame Prompting and Temporal Distilling (2023)

Monitoring Animal Pose Dynamics from Monocular Images (2022)

Image-based Complementary Item Recommendation Algorithm (2022)

Auto Design: ML-based Furniture Layout Recommendation Algorithm (2022)

EDUCATION

University of North Carolina at Chapel Hill

Ph.D., Computer Science

Chapel Hill, NC

2012 – 2017

Shanghai Jiao Tong University

B.E., Software Engineering

Shanghai, China

2008 – 2012